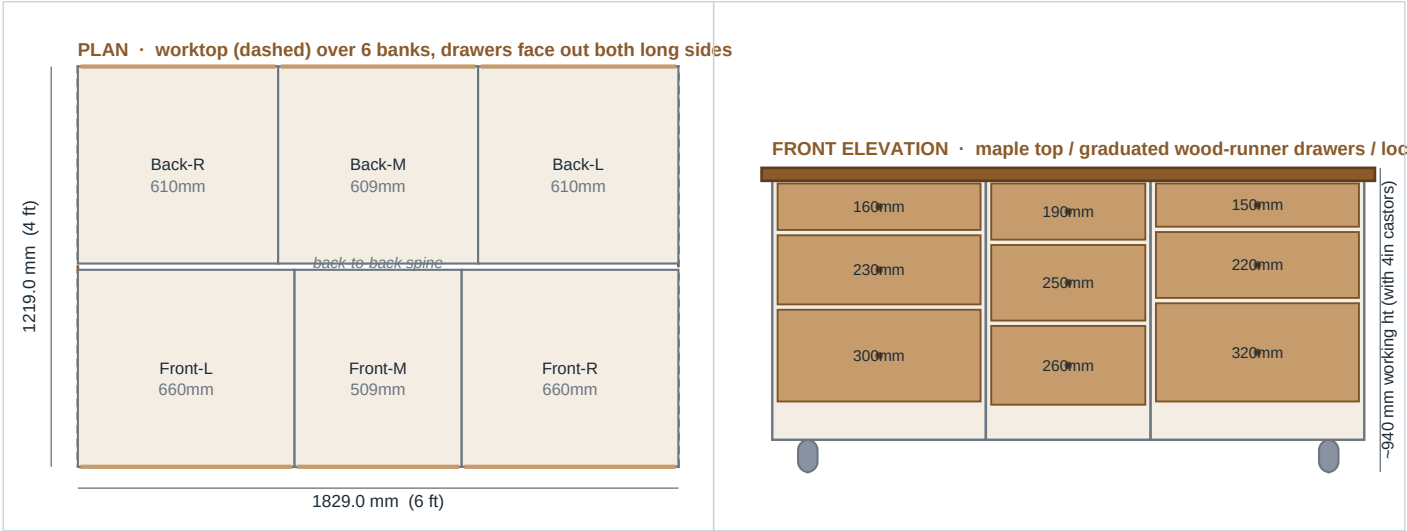


Mobile Storage Workbench

Double-sided mobile island · storage-focused · 4 ft × 6 ft

OVERALL	1829.0 × 1219.0 mm (6 × 4 ft)	WORKING HT	≈915–940 mm (on castors)
DRAWERS	18 · graduated · 3 tiers × 6 banks	RUNNERS	wood-on-wood (no metal slides)
WORKTOP	40.0 mm laminated hard maple	CARCASS	18.0 mm birch plywood
MOBILITY	4 × heavy-duty locking castors	EST. COST	\$2,564 materials



Construction & specification

Carcass	18.0 mm birch plywood; sides house the bottom in a 9.0 mm dado.
Drawer boxes	12.0 mm ply, dovetailed corners (tails on the sides — can't pull off), 6.0 mm captured bottom.
Drawer runners	Hardwood side runners screwed to the carcass; drawer sides grooved to ride them. No metal slides, no slide drilling.
Worktop	40.0 mm laminated hard maple, one continuous slab flush to the 4×6 footprint.
Mobility	4 heavy-duty locking swivel castors (rate ≥200 kg each) on corner blocks / base rail.
Stability	Low, wide, heavy double-sided island; wood runners aren't full-extension, so no wall anti-tip (deliberate).

Bank schedule

Bank	Width (mm)	Drawer fronts, top → bottom (mm)
Front-L	660.0	160.0 · 230.0 · 300.0 (sum 690.0)
Front-M	509.0	190.0 · 250.0 · 260.0 (sum 700.0)
Front-R	660.0	150.0 · 220.0 · 320.0 (sum 690.0)
Back-R	610.0	200.0 · 240.0 · 250.0 (sum 690.0)
Back-M	609.0	150.0 · 210.0 · 330.0 (sum 690.0)
Back-L	610.0	180.0 · 240.0 · 270.0 (sum 690.0)

Cut-list summary (full per-part list in cutlist.csv)

Part	Qty	Typical size (mm)	Material
Drawer box side	36	569.0 x 110.0 x 12.0 (+14 sizes)	drawer box
Side	12	800.0 x 600.0 x 18.0	sheet
Bottom	6	473.0 x 594.0 x 18.0 (+3 sizes)	sheet
Drawer box bottom	18	484.0 x 569.0 x 6.0 (+3 sizes)	back panel
Top stretcher	12	473.0 x 80.0 x 18.0 (+3 sizes)	sheet
Back	6	782.0 x 473.0 x 6.0 (+3 sizes)	back panel
Drawer box front/back	36	460.0 x 150.0 x 12.0 (+16 sizes)	drawer box
Run countertop	1	1829.0 x 1219.0 x 40.0	countertop
Drawer front	18	503.0 x 190.0 x 18.0 (+16 sizes)	door/front
Drawer wood runner	36	569.0 x 30.0 x 18.0	solid panel

Parts	181 (115 unique)	Sheet goods	~30.1 m² (324 ft²)
Hardwood runners	36 strips	Edge banding	50.0 m
Drilling	108 holes (all wood-runner screws — 0 slide holes)		
Est. build time	~33 h (advanced)	Est. materials	\$2,563.68

Generated by the Woodworking AI pipeline (validator + geometry critic passed, 0 interferences). Full reports alongside: cutlist · drilling · joinery · cutlayout.dxf.
Dimensions overall in mm unless noted.

Materials to buy (bill of materials)

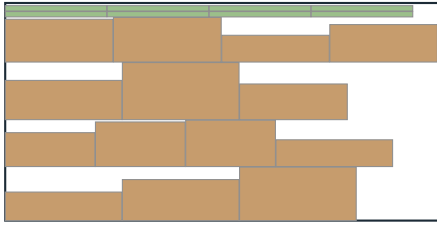
Category	Item	Qty	Unit	Buy cost
Sheet goods	Birch plywood — 12.0 mm, 2440.0 × 1220.0 mm sheet	3	sheet	
Sheet goods	Birch plywood — 18.0 mm, 2440.0 × 1220.0 mm sheet	5	sheet	
Sheet goods	Birch plywood — 6.0 mm, 2440.0 × 1220.0 mm sheet	4	sheet	
Worktop	Hard maple worktop slab — glue-up ≈1830.0 × 1220.0 × 40.0 mm	1	slab	
Lumber	Hardwood for drawer runners (Birch sheet)	4.7	bd ft	\$32.34
Hardware	Drawer pull	18	ea	\$63.00
Hardware	Assembly screw 4×35	24	ea	\$2.40
Hardware	Back panel screw 4×16	60	ea	—
Hardware	Edge banding	6	ea	—
Edge banding	Edge banding — sheet	50.0	m	\$74.94
Consumable	Wood glue (PVA)			\$96.00
Consumable	Sandpaper (assorted grits)			\$5.40

Sheet count is from the nesting layout (12 plywood sheets + 1 maple slab, 78% yield). Labour (~\$1,131, ~17 h) and consumables are shown for a full quote; materials + hardware to buy total roughly \$1,433.

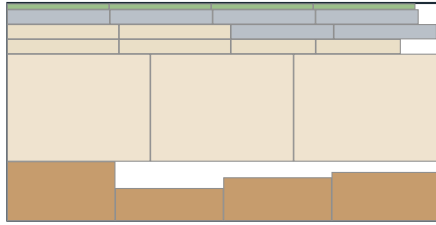
Cut sheets (nesting layout on 2440.0 × 1220.0 mm stock)

■ Carcass side/bottom
 ■ Drawer front
 ■ Drawer box
 ■ Wood runner
 ■ Back panel
 ■ Worktop

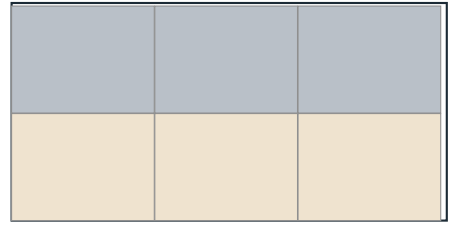
Sheet 1 — 18mm birch ply 68% used · 22 parts



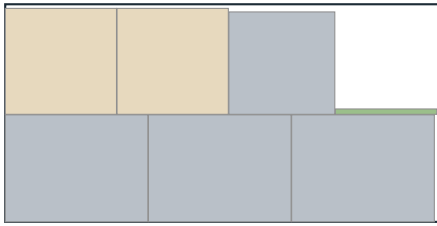
Sheet 2 — 18mm birch ply 90% used · 23 parts



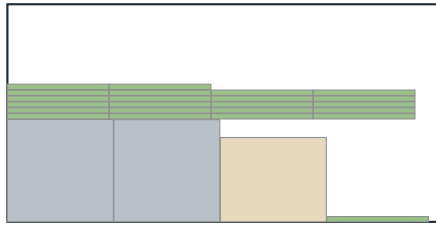
Sheet 3 — 18mm birch ply 97% used · 6 parts



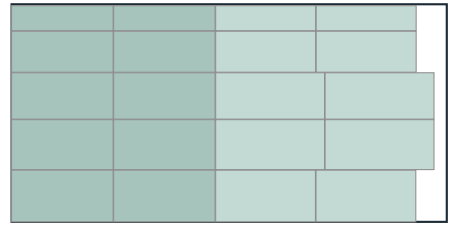
Sheet 4 — 18mm birch ply 85% used · 7 parts



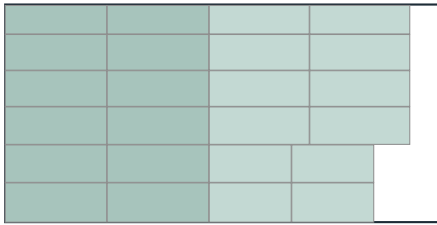
Sheet 5 — 18mm birch ply 46% used · 26 parts



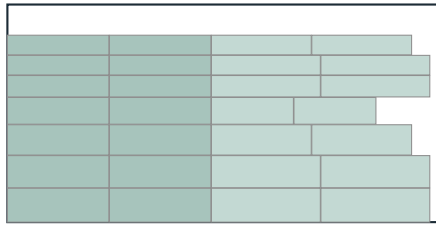
Sheet 6 — 12mm birch ply 93% used · 20 parts



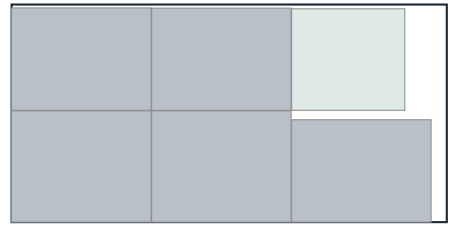
Sheet 7 — 12mm birch ply 88% used · 24 parts



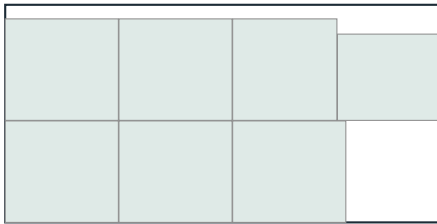
Sheet 8 — 12mm birch ply 79% used · 28 parts



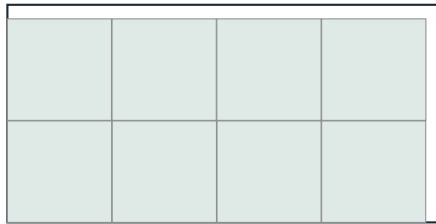
Sheet 9 — 6mm birch ply 90% used · 6 parts



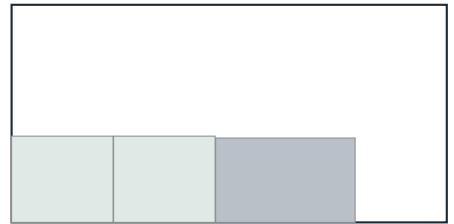
Sheet 10 — 6mm birch ply 81% used · 7 parts



Sheet 11 — 6mm birch ply 89% used · 8 parts



Sheet 12 — 6mm birch ply 31% used · 3 parts



Sheet 13 — 40mm maple slab 75% used · 1 parts

